MUHAMMAD **FAREED**

Lahore, Pakistan | mfareedkhan012@gmail.com | +92 320 4496708

Website | Google Scholar | GitHub | LinkedIn | **ORCID:** 0009-0005-9509-2108

Research Interests

Multimodal and trustworthy AI for healthcare; medical vision-language models and clinical report generation; multimodal medical-image and clinical-data learning; robust and uncertainty-aware medical AI; parameter-efficient adaptation and domain generalization.

Education

Master of Science (MS) in Data Science

Feb 2024 - Apr 2026

Riphah International University, Islamabad — **Lahore Campus**

CGPA: 4.0/4.0 (87.36%) | **First position** in the graduating cohort | **Merit Scholarship** 2024-2026

- **Thesis:** Retrieval-Augmented, Parameter-Efficient Fine-Tuning of Vision–Language Model for Chest X-ray Report Generation. | **Supervisor:** Dr. Jamal Uddin.
- **Selected coursework:** Machine Learning; Deep Learning; Statistical & Mathematical Methods for Data Science; Research Methods; Big Data Analytics; Advanced Database Systems

Bachelor of Education (B.Ed) Secondary (1.5 Year)

Oct 2021 - Mar 2024

Virtual University of Pakistan — **Lahore, Pakistan**

- **Specialization** in Information & Communication Technology (ICT) in Education.

Bachelor of Science (Honors) in Mathematics

2012 - 2016

University of Education, Township Campus — **Lahore, Pakistan**

- **Selected coursework:** Linear Algebra; Probability; Mathematical Statistics; Numerical Methods.

Publications & Manuscripts | Google Scholar

Published

[1] **Fareed, M.**, Fatima, M., Uddin, J., Ahmed, A., & Sattar, M. A. (2025). "A Systematic Review of Ethical Considerations of Large Language Models in Healthcare and Medicine". *Frontiers in Digital Health.* | Paper/Code

Accepted for Publication

[2] **Fareed, M.**, Uddin, J., Sayyid, A. H., Ahmed, A., & Sattar, M. A. (2026). "CXRG-SVLM: A Parameter-Efficient Small Vision-Language Model for Multi-View Chest X-ray Report Generation in Resource-Constrained Environments". *Scientific Reports.*

Submitted and Under Review

[3] **Fareed, M.**, Uddin, J., Ahmed, A., Nauman, M. A., Khan, N., Billah, M., & Sattar, M. A. (2026). LSTM-Based Deep Learning Models in Healthcare and Medicine: A Systematic Review. *PLOS ONE.*

Manuscripts in Preparation

[4] Ahmed, A., Riaz, S., & **Fareed, M.** (2026). "PEFT-mBERT: Benchmarking a Transformer-Based Encoder via Low-Rank and Activation-Based Adaptation for Resource-Efficient Punjabi Sentiment Analysis". *Manuscript being revised for resubmission.*

[5] Mir, S. S., Uddin, J., & **Fareed, M.** (2026). "Hybrid CNN–Transformer Based Diabetic Retinopathy Detection with Explainable AI Using Multi-Dataset Fundus Images".

Research Experience

Graduate Researcher – MS Thesis

2024 - 2026

Riphah International University, Lahore | **Advised by:** Dr. Jamal Uddin

CXRG-SVLM: Parameter-Efficient Multi-View Chest X-ray Report Generation

- Developed and evaluated a **medical vision-language model for multi-view chest X-ray report generation**, combining a frozen RAD-DINO medical image encoder with a 4-bit Qwen2.5-3B language model and **rank-16 QLoRA adaptation**.
- Integrated frontal and lateral radiographs through **attention-based multi-view fusion**, projection, and compression into 32 visual tokens, jointly conditioning generation on **medical-image and clinical-indication representations**.

- Implemented parameter-efficient multimodal training with approximately **38.1M trainable parameters (1.91%)**, together with an auxiliary **contrastive visual-text alignment objective**.
- Evaluated generated clinical reports using **lexical, semantic, radiology-specific, and entity-level measures**, including BERTScore, RaTEScore, and RadGraph-F1.
- Conducted **failure-mode and generalization analysis**, examining clinically relevant omissions, unsupported findings, underrepresented abnormalities, and the need for external and expert clinical validation.
- Led implementation, experimentation, evaluation, and manuscript preparation; **first-author work accepted for publication in Scientific Reports, 2026.**

First-Author Healthcare AI Research

2025 - 2026

Trustworthy and Ethical Large Language Models in Healthcare | **Advised by**: Dr. Jamal Uddin, Dr. Muhammad Awais Sattar, & Dr. Madeeha Fatima

- Led a systematic review of **ethical and trustworthy AI issues in healthcare LLMs**, synthesizing 27 studies from 316 screened records.
- Examined **privacy, bias, transparency, accountability, human oversight, safety, and regulatory and technical safeguards** for responsible deployment of large language models in healthcare.
- Developed a four-layer framework covering **regulatory safeguards, technical safeguards, human oversight, and accountability**.
- Published as first author in **Frontiers in Digital Health (2025)**: A Systematic Review of Ethical Considerations of Large Language Models in Healthcare and Medicine.

Collaborative Research

2026

Medical image analysis and parameter-efficient NLP | **Advised by**: Dr. Jamal Uddin & Dr. Saman Riaz

- Contributed to research design, experimentation, and **explainable multi-dataset medical-image analysis** for diabetic-retinopathy detection using fundus images and a hybrid CNN-Transformer architecture.
- Contributed to research on **parameter-efficient multilingual BERT adaptation**, including research design, literature analysis, and manuscript development.

Selected Technical Projects | GitHub

Co-team lead, Institute of Advanced Engineering (IAE) technical component associated with 'Innovation-Driven Product Development Using AI'. Co-led three team projects completed in January 2026.

Roman Urdu Conversational Language Model | **Advised by**: IAE Technical Team *Jan 2026*

- Co-led development of a conversational language-model fine-tuning pipeline using **Llama-3.1-8B, Unsloth, and 4-bit QLoRA**.
- Performed instruction fine-tuning and qualitative evaluation of generated responses under limited-compute conditions.

Length-Controlled News Summarization | **Advised by**: IAE Technical Team *Jan 2026*

- Co-led **LoRA adaptation of 4-bit Mistral-7B-Instruct** for controlled text generation.
- Evaluated model outputs using ROUGE, BERTScore, length adherence, and qualitative inspection of unsupported generated content.

Multilingual Sentiment Detection | **Advised by**: IAE Technical Team *Jan 2026*

- Co-led efficient adaptation of **Gemma-3-270M** for multilingual text classification, exploring lightweight language-model adaptation.

Professional Experience

Nov 2026 - Present **Visiting Lecturer — Computer Vision** at Riphah International University, Lahore Campus.

- Teaching Computer Vision to BS-AI undergraduate students.

Sep 2026 - Present Working as a **Management Trainee Officer — AI/ML** at Excellence Delivered (ExD), Lahore.

- Backend AI engineering for CONCORD.AI: API/LLM integration in role-based document workflows.

Teaching Experience

2018 - Present Working as a **Science & Mathematics Teacher (PST, BPS-14)** at GES Attari Saroba, Lahore | School Education Department, Government of Punjab.

2022 - 2023 Worked as a **Digital Marketing and Web Development Trainer** at Digiversity Training Institute, Lahore.

2015 - 2017 Worked as a **Mathematics and Physics Instructor** at Leads Group of Colleges, Lahore.

- Taught Linear Algebra and Probability & Statistics to BS IT/CS students, and mathematics and physics at intermediate level.

2014 - 2015 Worked as a **Private Mathematics Tutor** for national & international students (O & A-Levels).

2012 - 2014 Worked as a **Science and Mathematics Teacher** at Yasin Public High School, Lahore.

Honors, Awards & Scholarships

- **Top 1% Merit-Based Scholarship,** Riphah International University — 50% scholarship throughout the MS Data Science program, 2024–2026.
- **Certificate of Merit Achievement,** Dean's Honor List — MS Data Science, Riphah School of Computing & Innovation.
- Selected for the inaugural **AI-Driven Enterprise Product Training & Development Program**, PMYP × Stanford Engineering CGOE, October 2025 — selection through project-based assessment, profile review, and interview.
- **Best Teaching Performance Award**, Progressive Education Network, Lahore, December 2024.
- **Merit-based laptop award**, CM Punjab Laptop Program, 2012.
- **Full merit scholarship**, Punjab College of Science, 2010–2012 — tuition and associated study costs for F.Sc. Pre-Engineering.
- **First position in school**, Secondary School Certificate in Science, 2010.

Research Presentations

Riphah School of Computing & Innovation, Riphah International University, Lahore.

9 Dec 2025 **Three Minute Thesis Competition** — International Conference on Computing & Innovation (ICCI 2025); presenter, certificate of participation.

Sep 2025 **Research Week** — Poster presentation of MS thesis research.

Technical Skills

Medical & Multimodal AI	Medical vision-language models; clinical report generation; multimodal fusion; medical image analysis; clinical-text conditioning; visual-text alignment
Efficient Adaptation	LoRA; QLoRA; PEFT; 4-bit quantization; transfer learning; Unsloth
Deep Learning	PyTorch; TensorFlow; Keras; Hugging Face Transformers; CNNs; Transformers; medical vision encoders
Trustworthy Clinical AI & Evaluation	BERTScore; RaTEScore; RadGraph-F1; ROUGE; METEOR; entity-level evaluation; failure analysis; generalization assessment

Selected Certifications & Teacher Trainings

Innovation-Driven Product Development Using AI	Stanford Engineering CGOE x PMYP Partner Program	Feb 2026	Certificate
Deep Learning Specialization (completed all five courses)	DeepLearning.AI	Dec 2025	Certificate
Data Science Ethics	University of Michigan	Sep 2025	Certificate
Generative AI Engineering & Fine-tuning Transformers	IBM	Jul 2025	Certificate
Four-Week English Language Proficiency Program	PECTAA, Lahore	Aug 2026	Certificate
Six-Week Higher-Order Thinking Skills Programme	PECTAA, Lahore	Dec 2025	Certificate
Four-Week Induction Training Programme	QAED, Lahore	Feb 2018	Certificate

Leadership & Community Service

2018 - 2020 Worked as **Vice President** at Tameer Kadah Foundation, Lahore.

- Helped organize mobility-aid distribution drives, free medical and screening camps, and monthly financial assistance and ration deliveries, including support for children with disabilities.

References

Dr. Jamal Uddin, Associate Professor; Head, Department of AI & Data Science, RSCI, Riphah International University, Lahore

jamal.din@riphah.edu.pk | MS thesis supervisor

Dr. Muhammad Yaseen, Associate Professor; Head, Department of Software Engineering, In-Charge, Graduate/Postgraduate Programs, RSCI, Riphah International University, Lahore

muhammad.yaseen@riphah.edu.pk | MS coursework instructor

Dr. Muhammad Awais Sattar, Postdoctoral Researcher, Department of Computer Science, Electrical & Space Engineering Luleå University of Technology, Sweden

muhammad.awais.sattar@associated.ltu.se | Research mentor, coauthor, and MS coursework instructor